

Household Fuel Price Risk and the Uneven Protective Value of Domestic Net Zero: Evidence from Westminster, London

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Abstract

Fuel poverty identifies current affordability hardship, but it does not by itself measure future household exposure to fossil-fuel price shocks. This paper develops the concept of household fuel price risk and applies it to Westminster, London, using property-level or property-archetype modelling. The empirical strategy estimates required energy demand, baseline energy burden, price-shock sensitivity, and the distributional effect of domestic low-carbon interventions. The central argument is that domestic net zero is not automatically a household risk-protection strategy. Retrofit, clean heat, solar PV, battery storage, smart tariffs, and flexibility reduce unequal exposure only when housing fabric, heating systems, tariff design, market reform, tenure, and local governance align.

Keywords: fuel poverty; energy burden; price risk; domestic retrofit; net zero; Westminster; energy justice

1 Introduction

Fuel poverty research has made household energy hardship visible, but the recent politics of domestic energy requires a further question. A household may not be classified as fuel poor at one point in time and yet remain highly exposed to the next fossil-fuel price shock; equally, a household may receive a low-carbon technology without receiving meaningful protection from volatile retail energy costs. This paper therefore reframes household energy inequality from a static problem of current fuel poverty into a dynamic problem of uneven exposure to future fossil-fuel price volatility.

This reframing is significant for three reasons. First, energy-price shocks are not merely temporary deviations from a normal affordability problem. They reveal which households have high required energy needs, thin residual incomes, limited ability to absorb bill increases, and little control over the building or tariff conditions that shape their exposure. Second, actual expenditure can understate hardship when households respond to high prices by underheating, self-rationing, or delaying payment. A risk framework must therefore start from required energy services and then ask how costs change under plausible price and policy scenarios. Third, domestic net zero is increasingly presented as a route to lower bills and improved resilience, but the household-protection effect is not automatic. It depends on whether technical change reduces required demand, changes the household's fuel-price exposure, and is accessible under the household's tenure, dwelling form, tariff options, and local governance constraints.

Existing studies provide important building blocks but do not fully answer this paper's question. Fuel poverty indicators identify current need and support policy targeting, while energy vulnerability research shows that hardship is produced through income, housing quality, health, tenure relations, social support, and institutional access. Recent work on energy price resilience, energy poverty premiums, underheating, and crisis-period demand reduction moves closer to household price-risk analysis. Retrofit and low-carbon technology studies also show that fabric

improvements, heat pumps, PV, batteries, and flexibility can alter energy demand and bills. Yet these literatures often remain separated: fuel poverty metrics are usually current-condition indicators; retrofit studies often focus on average energy or bill effects; low-carbon adoption research examines who installs technologies more than who receives price-risk protection; and price-shock studies rarely connect future volatility to local housing governance and intervention feasibility at property or small-area scale.

The gap is especially important in the United Kingdom because decarbonisation and de-risking can diverge. Electrification can reduce direct household gas use and emissions, but it may not reduce household fuel price risk if electricity prices remain high relative to gas or continue to transmit gas-market shocks through wholesale and retail bill structures. Similarly, PV, batteries, and smart tariffs may hedge electricity-price exposure for some households while excluding others because of roof access, leasehold governance, smart-meter access, digital capability, load flexibility, or capital cost. The policy question is therefore not whether a measure is low carbon in engineering terms, but whether it reduces upper-tail household burden risk for the households most exposed to price volatility.

Westminster is an analytically revealing case rather than a merely convenient location. High average wealth can obscure fuel-price vulnerability among households facing high housing costs, inefficient or hard-to-retrofit dwellings, private rental constraints, leasehold governance, heritage restrictions, and limited access to low-carbon assets. Dense flats, social housing blocks, private rented weak-fabric dwellings, electric-heated inefficient flats, large inefficient gas-heated homes, and conservation-area properties create different routes into exposure and different routes out of it. These conditions make Westminster a strong test of whether domestic net zero can operate as household price-risk protection, and for whom.

The paper’s starting point is therefore not that domestic net zero measures are inherently protective. A dwelling can become lower carbon without becoming less exposed to household price risk if the intervention leaves required heat demand high, shifts the household onto a more expensive fuel, excludes the occupant from favourable tariffs, or depends on building-level decisions outside the household’s control. Conversely, the same intervention may be protective where fabric improvement, heating-system efficiency, tariff design, finance, and governance align. The key distributional question is who receives this protection and who remains exposed.

The central research question is: to what extent, and under what housing, market, and governance conditions, do domestic net zero interventions reduce unequal household exposure to fossil-fuel price volatility in Westminster?

The paper addresses three linked questions:

1. Which household-dwelling types face the largest required energy burden increase under fuel price shocks?
2. How do income, housing fabric, heating system, tenure, and vulnerability jointly produce fuel price risk?
3. How much risk reduction do retrofit, clean heat, PV, battery storage, smart tariffs, and integrated packages deliver, and for whom?

The contribution is fourfold. Conceptually, the paper distinguishes fuel poverty as current hardship, energy burden as a cost-to-income relation, and household fuel price risk as future-facing exposure, sensitivity, and limited adaptive capacity under price-shock scenarios. Methodologically, it develops a transparent income–cost and scenario framework that links required energy demand, local residual-income distributions, fuel-price shocks, intervention feasibility, and upper-tail burden risk. Empirically, it applies this framework to Westminster’s heterogeneous housing stock, where high-cost urban housing, tenure governance, flats, heritage constraints, and social housing delivery pathways are central mechanisms rather than contextual details. For policy,

it evaluates domestic net zero interventions as conditional de-risking tools, not as automatic bill-reduction or emissions-reduction proxies.

The modelling strategy follows directly from this argument. The baseline layer estimates required energy burden and the probability of crossing a burden threshold, because these measures identify where current cost and income conditions make exposure plausible even without observing each occupant’s income. The price-shock layer then varies fuel prices, standing charges, tariff structures, and mitigation to estimate how exposure changes under future conditions. The intervention layer finally asks whether fabric retrofit, clean heat, PV, storage, smart tariffs, or integrated packages reduce upper-tail burden risk once access constraints are included. The equations are therefore not a detached technical apparatus; they are the operational form of the paper’s central claim that current hardship, future exposure, and conditional protection must be measured separately before their distribution can be compared.

2 Conceptual Framework

The paper separates three concepts that are often conflated. Fuel poverty identifies present affordability hardship under current prices and required energy needs. Energy burden measures required or actual energy cost relative to residual household income. Household fuel price risk captures future exposure, sensitivity, and limited adaptive capacity under price-shock scenarios. The first is a condition, the second is a ratio, and the third is a forward-looking risk relation.

$$\text{Household fuel price risk} = \text{exposure} + \text{sensitivity} - \text{adaptive capacity}. \quad (1)$$

This expression is a conceptual guide rather than a claim that risk is literally additive in all empirical settings. Its purpose is to keep the three causal channels visible. Exposure refers to the household’s dependence on fuels, tariffs, standing charges, and required energy demand that are vulnerable to price shocks. Sensitivity refers to how strongly a price change affects household welfare, largely through residual income, required energy services, health needs, household composition, and debt or arrears. Adaptive capacity refers to the household’s ability to avoid, absorb, or reduce the shock through savings, tariff switching, demand flexibility, fabric improvement, heating-system change, PV or storage access, public support, landlord action, or building-level governance.

This distinction matters because the same observed energy bill can imply different risks. A low bill may indicate an efficient dwelling, but it may also indicate underheating or self-rationing. A high required energy cost may be manageable for a high-income household, but destabilising for a household with low after-housing-cost income. A low-carbon heating system may reduce direct gas exposure, but it can increase sensitivity to electricity prices if the dwelling remains inefficient or if the household cannot access an appropriate tariff. Household fuel price risk therefore cannot be read from EPC rating, current expenditure, or technology adoption alone.

The implemented baseline burden indicators should therefore be read as current-price affordability measures, not as the final fuel-price-risk outcome. They provide the income–cost foundation for later shock scenarios, but they do not themselves represent future exposure unless fuel-price, tariff, and intervention counterfactuals are introduced.

In the empirical work, this distinction is operationalised through two related but non-equivalent burden measures. The first is a deterministic baseline required-energy-burden ratio: SAP-derived required energy cost divided by MSOA after-housing-cost disposable income. The second is a probabilistic threshold measure: the probability that a household drawn from the property’s LSOA income distribution would face a required energy burden above 10 percent. The former is intuitive as a cost-to-income ratio; the latter is better suited to identifying local income-tail exposure. Neither measure is an official fuel-poverty classification, and neither should be read as future household fuel price risk before price shocks are modelled.

The use of required energy costs is central to the concept. Required demand asks what a household would need to spend to secure adequate domestic energy services, rather than what it actually spends after coping strategies have already occurred. This is analytically preferable where high prices may lead households to reduce heating, hot water, cooking, lighting, or appliance use below acceptable levels. A risk framework based only on actual expenditure would risk misclassifying households that appear inexpensive because they are already sacrificing comfort or welfare.

The probabilistic burden measure is also conceptually necessary. Westminster’s property records do not reveal the income of the current occupant, and a single MSOA income point estimate would hide lower-tail households living within otherwise affluent areas. Drawing from constrained LSOA income distributions does not claim to observe each household. Instead, it estimates how likely a property with a given required cost would become high-burden if occupied by households drawn from the local residual-income distribution. This is why the model reports both deterministic burden intensity and probabilistic threshold exposure.

This matters for interpretation. A top-decile modelled burden-risk group identifies the high end of Westminster’s current required-cost and income-distribution assumptions. It does not mean that every property in that group is occupied by a verified fuel-poor household, nor that all other properties are protected. Instead, it marks where required energy demand, local income sensitivity, and dwelling characteristics make high burden more probable under current prices. The future-facing risk analysis must then ask how this baseline exposure changes when gas prices, electricity prices, standing charges, tariff access, and policy mitigations are varied.

The conceptual framework also clarifies how domestic net zero interventions should be interpreted. Fabric retrofit primarily de-risks by lowering required heat demand, which reduces exposure to any future unit price. Clean heat primarily changes the fuel vector and system efficiency; its risk effect depends on the electricity-gas price ratio, seasonal performance, fabric condition, and tariff access. PV and batteries hedge electricity-price exposure only where roof rights, dwelling form, tenure, capital access, storage economics, and export or time-of-use tariffs permit the household to capture the benefit. Smart tariffs and flexibility can reduce costs for households able to shift demand, but can also reproduce inequality where digital access, smart-meter access, automation, or load flexibility are uneven. Bill support, standing-charge reform, and social tariffs reduce sensitivity directly, but do not necessarily reduce physical exposure.

This is why decarbonisation is not the same as de-risking. Electrification can reduce emissions before it reduces household price risk where electricity prices remain linked to gas through wholesale market design or where electricity remains expensive relative to gas at the retail level. Conversely, an intervention that is modest in carbon terms may still be protective if it reduces upper-tail burden risk for households with little adaptive capacity. The protective value of heat pumps, PV, storage, smart tariffs, and flexibility therefore depends on the electricity-gas price ratio, tariff access, capital access, dwelling form, tenure, and governance.

The framework implies a specific empirical test. A satisfactory model must identify baseline required-cost exposure, represent income sensitivity without pretending to know each occupant’s income, expose households to fuel-price and tariff scenarios, and then estimate whether intervention packages reduce upper-tail burden risk after feasibility constraints are applied. This is why the paper uses burden ratios, threshold-crossing probabilities, expected shortfall-style affordability gaps, Fuel Price-at-Risk, and de-risking dividends. Together, these measures are sufficient for the paper’s purpose because they answer the distributional question directly: not whether an intervention improves the average dwelling, but whether it reduces the probability and severity of unaffordable required energy costs for the households and dwelling types most exposed to shocks.

3 Case and Data

The empirical case is Westminster, London. The preferred unit of analysis is the property or household-dwelling pair. Where household microdata are unavailable, the study uses EPC property records combined with LSOA/MSOA income and deprivation proxies to assign synthetic households or archetypes.

Westminster is not treated as a generic affluent borough. Its relevance lies in the coexistence of high aggregate wealth with residual-income pressure, older and heterogeneous housing, private renting, leasehold flats, conservation and heritage constraints, and social housing retrofit pathways. These features are not background descriptors; they shape exposure, sensitivity, and adaptive capacity. In a high-cost city, households can face fuel-price vulnerability through the combination of required heat demand and constrained residual income even where area-level averages appear comfortable.

The current enriched property master contains 124,980 properties across 126 LSOAs and 26 MSOAs. It combines EPC attributes, spatial identifiers, MSOA after-housing-cost income, LSOA income deprivation, official LSOA fuel poverty rates, and LSOA income-distribution outputs from the latest verified pipeline.

The spatial outputs use the 123 Westminster LSOAs available in the current GIS boundary file. The three data LSOAs without matching polygons, containing 44 properties, are retained in all non-spatial analysis but omitted from the map panels. This boundary should be kept explicit when interpreting the spatial concentration results in Figure 6.

The latest EPC enrichment step links certificate records to the Westminster property master by UPRN, applies consistent UPRN normalisation in both merge and QA stages, and retains one selected EPC certificate per property using the latest lodgement or inspection date. The audited enriched stock preserves the master row count and reports zero unmatched master rows after the corrected linkage. EPC-derived tenure is used only as a proxy for tenure-related governance and adaptive capacity; the corrected analysis version classifies the stock as 58,744 private rented properties, 43,686 owner-occupied properties, 20,066 social rented properties, and 2,484 unknown-tenure properties.

The analysis uses required energy demand rather than only observed expenditure, because actual consumption may understate unmet need through underheating or self-rationing.

The resulting dataset is therefore best understood as a property-level modelling backbone for current required-burden exposure and future scenario analysis. It is not a direct observation of each household’s income, consumption, or current tenure contract. This boundary is important: the modelling design is intended to reveal which property–income–governance combinations are likely to be exposed, not to certify household-level fuel-poverty status.

4 Methods

The empirical strategy is a property-level income–cost and scenario model. It links EPC-derived dwelling characteristics, synthetic LSOA income distributions, required energy costs, price-shock scenarios, domestic net zero intervention packages, and spatial-distributional outputs. The unit of analysis is property p in LSOA l and MSOA m . Where observed household incomes are unavailable, the model estimates the probability that a household drawn from the local residual-income distribution would face a given required-energy burden. This design is deliberately probabilistic: it does not claim to observe the income of the current occupant of each property.

The equations are chosen to match the conceptual problem rather than to maximise statistical complexity. The paper needs a model that can do four things transparently: estimate required energy affordability under current conditions; expose the same property-income combinations to future fuel-price shocks; distinguish average burden from upper-tail burden risk; and compare intervention packages only after technical, tenure, tariff, and governance constraints are repre-

sented. A property-level scenario model is appropriate because the key mechanisms operate at different scales: dwelling fabric and heating systems are property-level, income distributions are small-area approximations, tariffs and fuel prices are market-level, and feasibility constraints are mediated by tenure, leasehold, heritage, and local delivery. The model is therefore not a prediction of exact bills for named households. It is a structured counterfactual framework for asking which property–income–governance combinations are most exposed and which interventions reduce that exposure.

This approach is also deliberately modular. The income–cost engine establishes the baseline denominator and required-cost numerator. The price-shock engine changes unit prices, standing charges, and mitigation while holding the logic of required demand explicit. The intervention engine then changes demand, fuel structure, tariff access, and feasibility. Because each layer modifies a different part of the same burden equation, the resulting de-risking estimates remain interpretable: a lower risk score can be traced to lower required demand, lower unit-price exposure, improved tariff access, policy support, or greater feasibility. This is why the equations are sufficient for the paper’s purpose even though they do not observe all household behaviours directly.

4.1 Income–Cost Engine

The income–cost engine answers the first empirical problem: how can current required-burden exposure be estimated when the dataset contains properties rather than verified household incomes? A single average income would suppress Westminster’s local lower tail, while assigning one synthetic income to each property would imply false precision. The constrained lognormal specification provides a middle route. It keeps MSOA after-housing-cost income as an anchor, allows LSOA-level lower-tail differentiation through deprivation information, and returns probabilities rather than definitive household labels.

The implemented income layer estimates one constrained lognormal after-housing-cost income distribution for each LSOA:

$$Y_l \sim \text{Lognormal}(\mu_l, \sigma_l^2), \quad (2)$$

where Y_l denotes annual residual household income. The distribution is parameterised by its mean $\bar{Y}_l$:

$$\mu_l = \ln(\bar{Y}_l) - \frac{1}{2}\sigma_l^2. \quad (3)$$

For each MSOA, the LSOA parameters are jointly estimated so that the property-weighted mean remains anchored to the MSOA after-housing-cost income, while the lower tail is differentiated by the LSOA income-deprivation signal. For a given regularisation parameter λ , the objective is:

$$\begin{aligned} \mathcal{L}_m(\lambda) = & \left[\ln \left(\frac{\sum_{l \in m} N_l \bar{Y}_l}{\sum_{l \in m} N_l} \right) - \ln(\bar{Y}_m) \right]^2 \\ & + \lambda \sum_{l \in m} w_l [F_l(z) - d_l]^2, \end{aligned} \quad (4)$$

where N_l is the number of EPC property records in LSOA l , $w_l = N_l / \sum_{l \in m} N_l$, $\bar{Y}_m$ is MSOA residual income, $z = 19,009$ is the income threshold used in the current module, $F_l(\cdot)$ is the LSOA lognormal cumulative distribution function, and d_l is the LSOA income-deprivation score. The current implementation constrains $\sigma_l \in [0.10, 1.40]$ and searches a log-spaced grid of 100 λ values from 10^{-4} to 10^{-1} . Official LSOA fuel-poverty rates are used to select λ by minimising property-weighted RMSE, with weighted MAE and then the smaller λ as tie-breakers; they are not entered directly as the optimisation target.

Required energy cost is derived from SAP and floor area rather than from observed consumption, because actual expenditure may be suppressed by underheating or self-rationing. For

property p , the SAP energy cost factor is:

$$\text{ECF}_p = \begin{cases} (100 - \text{SAP}_p)/16.21, & \text{SAP}_p \geq 43.265, \\ 10^{(108.8 - \text{SAP}_p)/120.5}, & \text{SAP}_p < 43.265. \end{cases} \quad (5)$$

The baseline required annual energy cost is then:

$$C_{p,0} = \text{ECF}_p \frac{A_p + 45}{0.2}, \quad (6)$$

where A_p is total floor area. Baseline required energy burden for a household with income Y_l is:

$$B_{p,0}(Y_l) = \frac{C_{p,0}}{Y_l}. \quad (7)$$

The current module therefore estimates burden-threshold probabilities:

$$P_{p,0}(\tau) = \Pr(B_{p,0}(Y_l) > \tau) = F_l\left(\frac{C_{p,0}}{\tau}\right). \quad (8)$$

The implemented baseline output uses $\tau = 0.10$. For modelled LSOA fuel-poverty comparison, properties with SAP grades A, B, or C are assigned a property poverty score of zero; other properties use $P_{p,0}(0.10)$. This score is a baseline affordability input, not a complete measure of household fuel price risk.

The threshold probability is useful because it reports exposure to a policy-relevant affordability boundary without claiming that the boundary is the whole concept of fuel poverty. It also avoids reducing Westminster's mixed-income geography to area averages. A property can have moderate deterministic burden under the MSOA mean but still have a substantial probability of high burden for households in the local lower tail. Conversely, a property with high required cost may be less probabilistically exposed if local residual incomes are high. This distinction is central to the paper's question because price shocks will interact with both required demand and income sensitivity.

For robustness, the pipeline also reports a deterministic baseline burden ratio:

$$\tilde{B}_{p,0} = \frac{C_{p,0}}{\bar{Y}_m}, \quad (9)$$

where $\bar{Y}_m$ is the MSOA after-housing-cost income point estimate. This measure is easier to interpret as a cost-to-income ratio, but it does not model within-LSOA income dispersion and is not calibrated to the local fuel-poverty comparison.

4.2 EPC Enrichment and High-Risk Pathways

The latest explanatory layer enriches the Westminster property master with EPC certificate records using UPRN-based linkage. UPRNs are standardised before both merging and QA checks, so that numeric and string representations of the same property identifier do not generate false unmatched cases. Where multiple EPC certificates are available for a property, the selected record is the latest `LODGEMENT_DATETIME`; where this is unavailable, the latest `INSPECTION_DATE` is used, and remaining ties fall back to the first available row. EPC fields are added with an `EPC_` prefix, the original property master is not overwritten, and the row count and row order of the master are preserved.

The enrichment is followed by an explicit audit stage covering unmatched master records, duplicate UPRNs, added-field missingness, EPC rating distributions, lodgement-year distributions, and original versus corrected tenure groups. The EPC-derived tenure variable is treated as a proxy for tenure-related adaptive capacity rather than as observed legal tenure. The analysis

version, `EPC_TENURE_GROUP_FIXED`, classifies social rented records before private rented records to avoid misclassifying social rental descriptions as private rental cases.

The enriched stock is then transformed into a descriptive high-risk layer. Derived variables capture EPC performance, SAP-derived required energy cost, fabric weakness, electric heating exposure, gas connection, corrected tenure proxy, baseline required energy burden, and membership in modelled high-burden groups. The main current-condition groups are the top decile and top quintile of $P_{p,0}(0.10)$, denoted HR10 and HR20. Robustness groups use the top decile and top quintile of deterministic baseline required energy burden, denoted BR10 and BR20; an additional deterministic threshold group identifies properties with $\tilde{B}_{p,0} > 0.10$. These groups describe current modelled burden probability and required-burden intensity; they are not official fuel-poverty classifications and do not by themselves constitute the future-facing household fuel price risk measure.

The distinction between HR and BR groups is analytically useful rather than merely technical. HR groups identify properties whose required costs interact with the local LSOA income distribution to produce a high probability of crossing a 10 percent burden threshold. BR groups identify properties with high deterministic required burden relative to the MSOA point estimate of after-housing-cost income. Their overlap is therefore expected to be partial: the two approaches should identify a related high-burden tail, while also showing whether income-distribution assumptions change who is treated as exposed.

Finally, high-risk properties are assigned to mutually exclusive descriptive pathways using priority rules. The current typology distinguishes inefficient large owner-occupied gas houses, social rented estate or block risk, private rented weak-fabric gas flats, electric-heated inefficient flats or maisonettes, high-demand inefficient maisonettes, and a residual high-risk group. The pathway typology is not a causal model. Its purpose is to provide an interpretable bridge between property-level burden modelling and the later price-shock and intervention simulations, so that de-risking outcomes can be reported by mechanism and governance pathway rather than only for the aggregate stock.

The corresponding descriptive figures report pathway risk and contribution, EPC–floor-area sensitivity, tenure structure, electric exposure, spatial concentration, and HR–BR overlap (Figures 1–6). They are current-condition burden-exposure diagnostics rather than price-shock results.

4.3 Price-Shock Risk Metrics

The price-shock layer converts the baseline income–cost engine into forward-looking risk metrics. Let $s \in S$ denote a fuel-price scenario, including gas-price, electricity-price, combined-fuel, standing-charge, and policy-mitigation cases. Required cost under scenario s is written as:

$$C_{p,s} = \pi_s^g Q_p^g + \pi_s^e Q_p^e + SC_s^g I_p^g + SC_s^e I_p^e - M_{p,s}, \quad (10)$$

where Q_p^g and Q_p^e are required gas and electricity quantities or archetype-imputed equivalents, π_s^g and π_s^e are scenario fuel prices, SC_s^g and SC_s^e are standing charges, I_p^g and I_p^e indicate fuel connections, and $M_{p,s}$ captures bill support or other mitigation. The scenario burden is:

$$B_{p,s}(Y_l) = \frac{C_{p,s}}{Y_l}, \quad (11)$$

and the burden uplift is:

$$\Delta B_{p,s}(Y_l) = B_{p,s}(Y_l) - B_{p,0}(Y_l). \quad (12)$$

This cost equation is intentionally simple and explicit. It separates the components through which price risk reaches households: gas unit prices, electricity unit prices, fixed standing charges, fuel connections, and policy mitigation. That separation is necessary because different interventions act on different terms. Fabric retrofit reduces required quantities; heat pumps

change the gas-electricity mix; PV and batteries reduce grid electricity imports under some profiles; tariff reform changes prices or standing charges; and public support enters as mitigation. A more compact total-bill equation would obscure these mechanisms and make it harder to explain why an intervention protects one group but not another.

The main risk indicators are defined as distributional rather than average outcomes. The scenario-specific threshold-crossing probability is:

$$P_{p,s}(\tau) = F_l \left(\frac{C_{p,s}}{\tau} \right). \quad (13)$$

For scenario weights ω_s , the expected threshold-crossing risk is:

$$R_p(\tau) = \sum_{s \in S} \omega_s P_{p,s}(\tau). \quad (14)$$

An affordability gap-at-risk can be expressed as the expected shortfall above an acceptable burden threshold:

$$G_{p,s}(\tau) = E_{Y_l} [\max\{C_{p,s} - \tau Y_l, 0\}]. \quad (15)$$

Household Fuel Price-at-Risk is defined as a high quantile of the scenario burden distribution:

$$\text{HFPaR}_{p,\alpha} = \inf \left\{ b : \sum_{s \in S} \omega_s \mathbf{1}(B_{p,s} \leq b) \geq \alpha \right\}, \quad (16)$$

where α may be set to a policy-relevant upper-tail level such as 0.90 or 0.95. These metrics keep fuel poverty, energy burden, and fuel price risk analytically separate: the first concerns current hardship, the second is a cost-to-income ratio, and the third measures future exposure and sensitivity under price shocks.

The three risk metrics have different policy uses. Threshold-crossing probabilities show how many property-income combinations are likely to move beyond an affordability boundary. Gap-at-risk reports the monetary depth of exposure above that boundary, which matters for bill support or social tariff design. Household Fuel Price-at-Risk focuses on the upper tail of the scenario burden distribution, which is the relevant object when the policy concern is protection against shocks rather than average annual affordability. Reporting all three avoids a false choice between headcount, severity, and tail-risk interpretations.

4.4 Intervention Counterfactuals

Domestic net zero interventions are represented as counterfactual changes to required demand, fuel structure, tariff exposure, and mitigation access. Let k denote an intervention package, such as fabric retrofit, clean heat, PV, battery storage, smart tariffs, or an integrated package. The post-intervention scenario cost is:

$$C_{p,s,k} = a_{p,k} \tilde{C}_{p,s,k} + (1 - a_{p,k}) C_{p,s}, \quad (17)$$

where $a_{p,k} \in [0, 1]$ represents technical, tenure, leasehold, heritage, roof, finance, or tariff feasibility. This term is central to the paper's interpretation: an intervention produces household risk protection only where access and market conditions allow the technical change to translate into lower exposure.

The feasibility parameter is not a decorative adjustment. It is the formal expression of the paper's claim that domestic net zero is conditional household protection. Without $a_{p,k}$, the model would implicitly assume that every property can receive every intervention and that all occupants can capture the resulting benefit. That assumption would be especially misleading in Westminster, where flats, leasehold blocks, private renting, social housing delivery, heritage

constraints, roof rights, smart-meter access, and tariff eligibility all shape whether an engineering option becomes a household de-risking option.

The intervention-specific threshold risk and Fuel Price-at-Risk are:

$$P_{p,s,k}(\tau) = F_l \left(\frac{C_{p,s,k}}{\tau} \right), \quad \text{HFPaR}_{p,\alpha,k} = Q_\alpha (\{B_{p,s,k}\}_{s \in S}). \quad (18)$$

The de-risking dividend is the reduction in upper-tail burden risk:

$$D_{p,\alpha,k} = \text{HFPaR}_{p,\alpha,0} - \text{HFPaR}_{p,\alpha,k}. \quad (19)$$

Results will be stratified by income distribution, EPC rating, dwelling form, heating system, tenure, leasehold and heritage constraints, and LSOA/MSOA location. The empirical question is therefore not whether a low-carbon measure reduces emissions in engineering terms, but who receives measurable price-risk protection once required demand, tariffs, fuel-market pass-through, and local governance constraints are jointly considered.

The de-risking dividend is the bridge between the conceptual and empirical contributions. It does not ask whether an intervention is technically impressive on average. It asks whether the intervention reduces the upper-tail burden that households would face across price-shock scenarios. A positive dividend means that the package has converted at least part of the technical or policy change into lower household price-risk exposure. A small or uneven dividend indicates either that the intervention does not affect the relevant risk channel, or that feasibility, tariff access, price pass-through, or income sensitivity prevents households from capturing the protection.

5 Expected Outputs

The empirical outputs are designed to move from an implemented income–cost baseline to forward-looking price-risk and intervention comparisons. The latest verified modelling round already provides the first layer of this sequence: a property-level baseline required-energy-burden engine for Westminster, built from constrained LSOA income distributions and SAP-derived required energy costs. The run labelled 20260520_213323 covers 124,980 EPC property records across 126 LSOAs and 26 MSOAs. It preserves MSOA after-housing-cost income consistency while allowing LSOA differentiation mainly through income dispersion and lower-tail probability, rather than by claiming directly observed household incomes.

5.1 Implemented Baseline Outputs

The current outputs support three baseline products. First, the model estimates LSOA lognormal income parameters, including μ_l , σ_l , lower-tail probabilities below £19,009, and the selected MSOA-level regularisation parameter λ . In the latest verified run, the ratio of LSOA mean income to MSOA income is tightly anchored, ranging from 0.999650 to 1.000309, while σ_l ranges from 0.4192 to 1.4000. Only 3 of 126 LSOAs sit at the new upper bound, compared with 31 of 126 under the previous bound of 0.9. This indicates that within-MSOA differentiation is being carried primarily by dispersion and tail probability.

Second, the model produces property-level SAP-derived required energy cost and the probability that required energy burden exceeds a 10 percent threshold. This output should be interpreted as baseline required-energy affordability exposure, not as a complete household fuel price risk measure. It provides the denominator and cost foundation for later shock scenarios, while avoiding dependence on actual expenditure that may be suppressed by underheating or self-rationing.

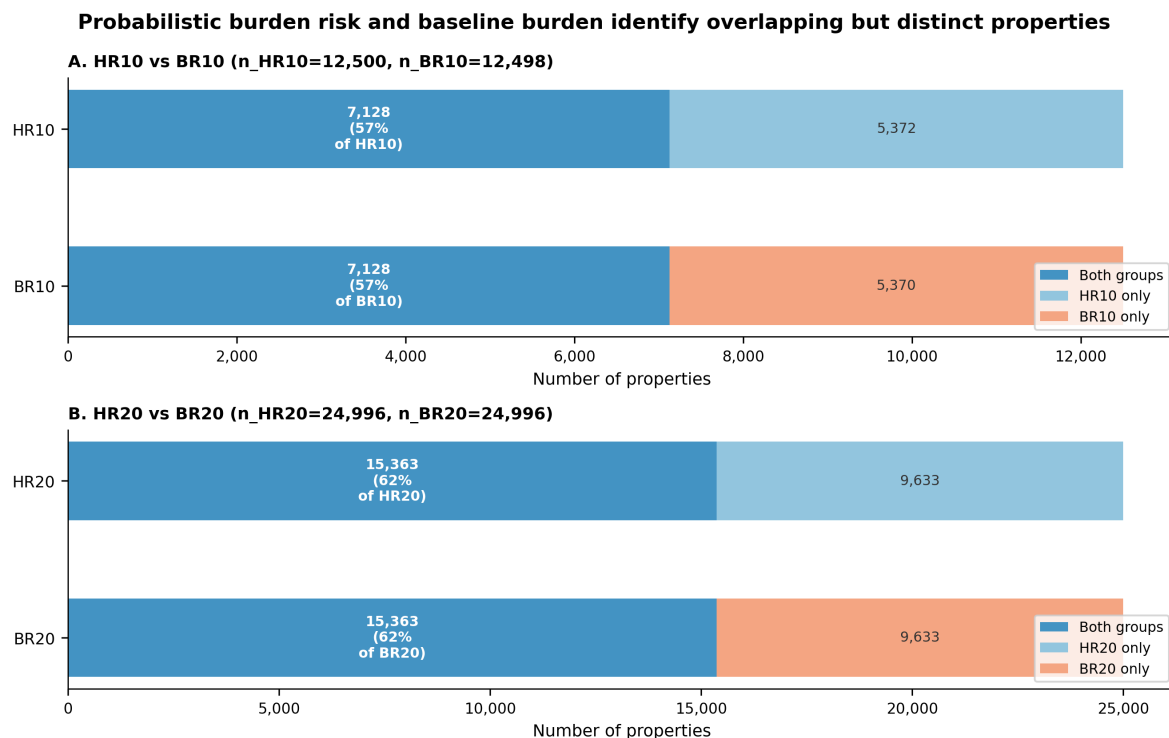
Third, the baseline module generates a modelled LSOA fuel-poverty comparison used for calibration diagnostics. The latest run has a mean modelled-minus-official LSOA fuel-poverty-rate error of -0.0174, a median error of -0.0270, and a worst absolute error of 0.133. This residual

gap is substantively useful rather than merely technical: it shows that income-tail matching alone does not reproduce official fuel-poverty geography, reinforcing the paper’s distinction between income deprivation, required energy burden, dwelling fabric, heating systems, tenure, and future price risk.

5.2 Implemented High-Risk Explanation Outputs

The latest high-risk explanation stage adds an audited EPC-enriched stock and a descriptive pathway typology. The enriched master retains 124,980 Westminster property records. After corrected UPRN standardisation, the QA workbook reports a complete match between the master properties and EPC certificate records, with zero unmatched master rows. The corrected EPC tenure proxy distributes the stock into 58,744 private rented properties, 43,686 owner-occupied properties, 20,066 social rented properties, and 2,484 unknown-tenure properties. These tenure categories are analytical proxies and should be interpreted as adaptive-capacity and governance indicators rather than as verified household tenure records.

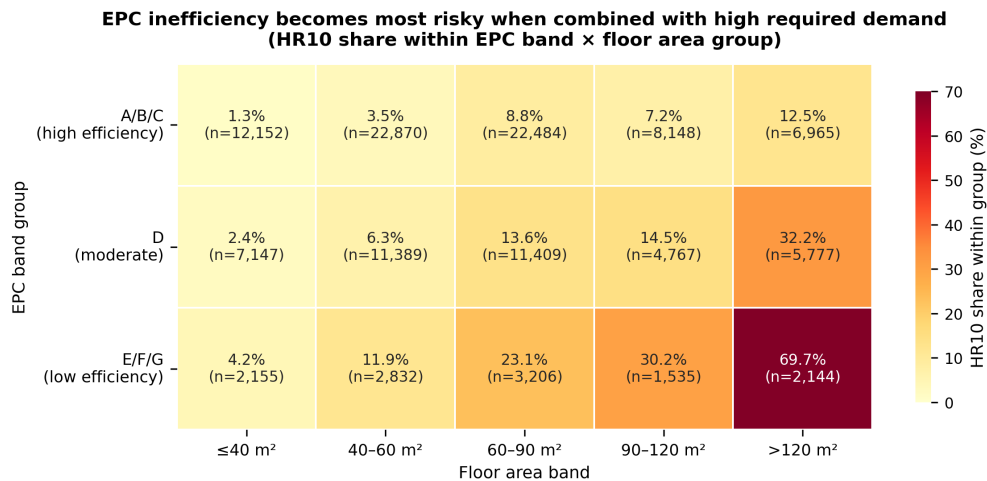
The implemented high-risk groups remain exploratory and model-based. HR10 contains the 12,500 properties in the top decile of the probability that required energy burden exceeds 10 percent; HR20 contains the 24,996 properties in the top quintile. The deterministic burden robustness groups contain 12,498 BR10 properties and 24,996 BR20 properties, while only 737 properties have a baseline required energy burden above 10 percent under the deterministic baseline measure. As Figure 1 shows, HR10 and BR10 overlap for 7,128 properties, or 57.0 percent of each group. This overlap is large enough to show that the probabilistic and deterministic indicators identify a related high-burden tail, but incomplete enough to justify reporting both.



Note: HR10/HR20 = top decile/quintile of burden_prob_gt_10 (modelled probability burden > 10% of income). BR10/BR20 = top decile/quintile of baseline_required_energy_burden (current required cost / income). The two metrics identify related but non-identical sets of high-risk properties. Neither is equivalent to official fuel poverty status.

Figure 1: Overlap between probabilistic high-risk groups (HR10/HR20) and deterministic baseline burden groups (BR10/BR20). The partial overlap shows that income-distribution exposure and deterministic burden intensity identify related but non-identical current high-burden sets.

The strongest descriptive signal in the current outputs is the interaction between dwelling size and energy performance. Figure 2 shows that EPC E–F–G properties above 120 m² have an HR10 share of 69.68 percent and an HR20 share of 85.21 percent, with a median baseline burden of 7.26 percent. E–F–G houses also show high within-type exposure, with an HR10 share of 56.08 percent. By contrast, E–F–G flats have a lower within-type HR10 share of 19.38 percent, but remain important because flats are numerous in the Westminster stock.



Note: HR10 = top-decile modelled burden probability group. EPC band and floor area are proxies for energy efficiency and required demand.

Figure 2: HR10 share by EPC band and floor-area group. The heatmap shows that low energy performance becomes most consequential where required demand is amplified by dwelling size; EPC band alone is therefore an incomplete risk signal.

Table 1: Current high-risk pathway typology from the latest explanatory run

Pathway	Properties	Stock	HR10	HR10 contrib.
Inefficient large owner-occupied gas houses	2,802	2.24%	50.25%	11.26%
Social rented estate/block risk	7,016	5.61%	22.66%	12.72%
Private rented weak-fabric gas flats	34,672	27.74%	4.18%	11.60%
Electric-heated inefficient flats/maisonettes	10,256	8.21%	13.83%	11.34%
High-demand inefficient maisonettes	2,220	1.78%	25.81%	4.58%
Other high-risk residual group	13,063	10.45%	46.40%	48.49%
Not high risk / comparison stock	54,951	43.97%	0.00%	0.00%

Table 1 should be read as a targeting and interpretation device, not as a final causal attribution model. It separates two policy-relevant rankings: the highest within-pathway risk is concentrated in inefficient large owner-occupied gas houses, high-demand inefficient maisonettes, and social rented estate/block risk, while the largest contributions to all HR10 properties come from social rented estate/block risk, private rented weak-fabric gas flats, electric-heated inefficient flats/maisonettes, and the residual high-risk group. Figure 3 visualises the same distinction between risk intensity and stock-scale contribution. The next empirical outputs should therefore report both rankings.

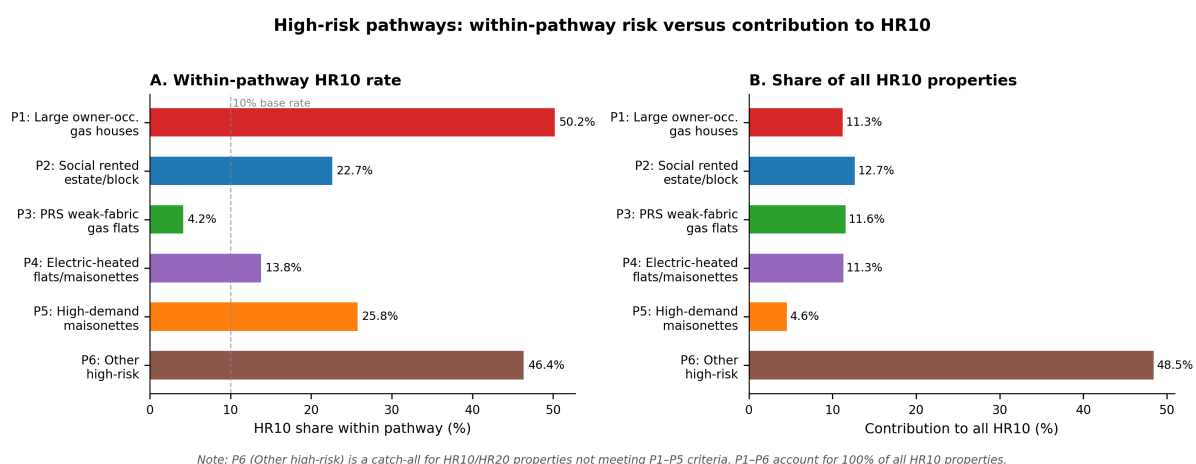


Figure 3: Within-pathway HR10 risk and contribution to all HR10 properties. The figure distinguishes intensity from scale: some pathways have high within-group exposure, while others contribute substantially because they are more numerous in the Westminster stock.

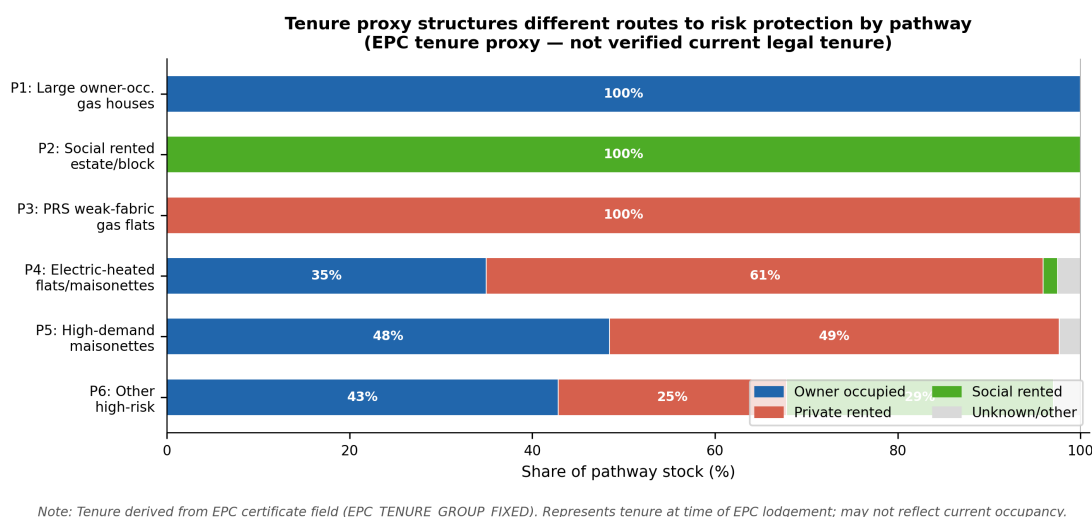


Figure 4: Tenure-proxy composition by high-risk pathway. The figure links each descriptive pathway to a different governance route, including owner-occupation, social housing delivery, private rented enforcement, and mixed residual cases.

Figure 4 then shows why these pathways imply different governance routes, rather than a single retrofit-delivery model. The electric-exposure comparison in Figure 5 isolates inefficient flats and maisonettes with EPC-recorded electricity exposure from non-electric low-EPC comparators and the remaining stock. Its purpose is not to claim that electrification causes high burden, but to show why household-facing electricity prices, fabric quality, and tariff access must be modelled before clean heat can be interpreted as price-risk protection.

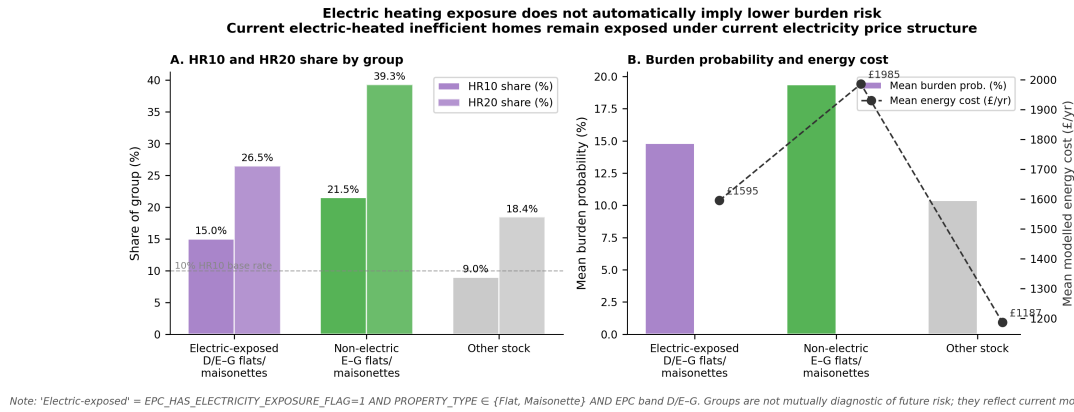


Figure 5: Current burden exposure among electric-exposed inefficient flats and maisonnettes compared with non-electric low-EPC flats and the rest of the stock. The figure supports the paper's caution that electricity exposure and electrification are not automatically household de-risking mechanisms under current price structures.

The spatial outputs in Figure 6 show that Westminster's current high-burden exposure is geographically uneven and mechanism-specific. They should be interpreted as LSOA-level descriptive maps of current HR10 concentration and dominant HR10 pathway, not as direct maps of future price-shock risk.

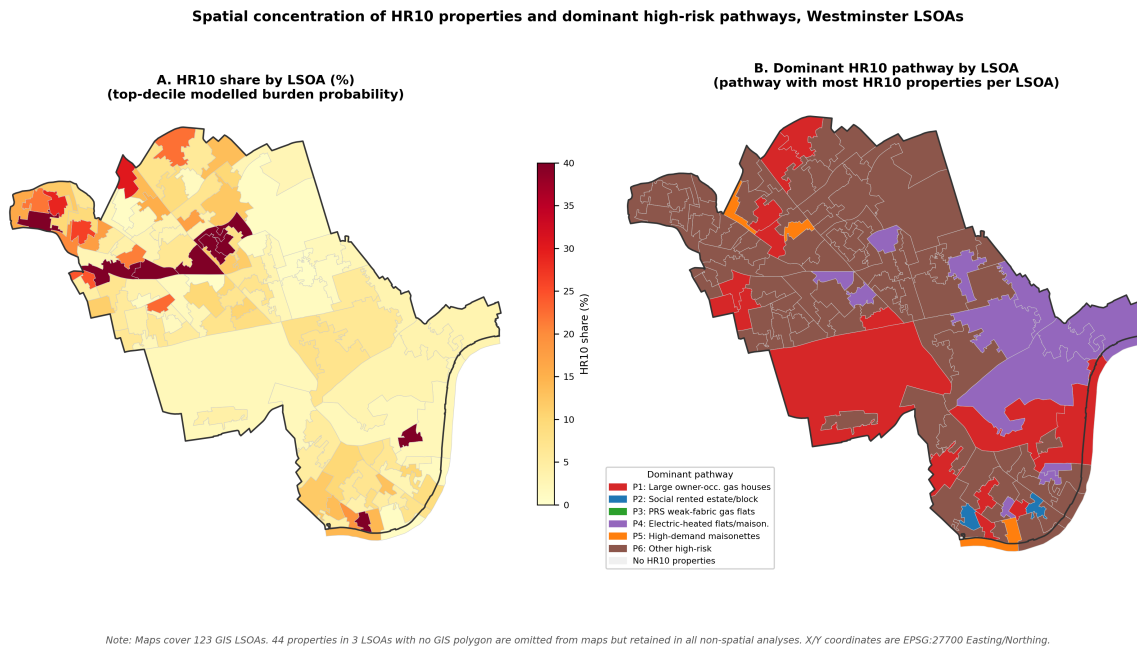


Figure 6: Spatial concentration of current modelled HR10 exposure in Westminster. Panel A maps the LSOA share of properties in HR10; Panel B maps the dominant high-risk pathway among HR10 properties in each LSOA. The maps use the 123 LSOAs available in the GIS boundary file.

The residual group is especially important for the next stage. It is too large to remain a black-box category once the paper moves from current exposure into price-shock risk. The planned analysis should separate income-driven residual cases, spatial-deprivation concentrations, mixed housing-and-income mechanisms, and unusual dwelling profiles. This will help distinguish households for whom technical retrofit is the central risk-reduction route from those for whom

social tariffs, standing-charge reform, income support, or local assistance may be more relevant.

5.3 Risk and Intervention Outputs

The next empirical layer will convert the baseline engine into price-shock outputs. These will include LSOA/MSOA maps, property-archetype summaries, distributional plots, and threshold-crossing estimates under gas-price, electricity-price, combined-fuel, standing-charge, and policy-mitigation scenarios. The paper will report not only average burden changes, but also upper-tail risk, burden uplift, and the probability that households cross policy-relevant burden thresholds.

A central output is the de-risking dividend:

$$\text{De-risking dividend} = \text{Fuel Price-at-Risk}_{\text{before}} - \text{Fuel Price-at-Risk}_{\text{after}}. \quad (20)$$

This metric will be estimated by intervention package and household-dwelling condition, including fabric retrofit, clean heat, PV, battery storage, smart tariffs, and integrated packages. Its interpretation will remain conditional: an intervention produces household price-risk protection only where technical feasibility, tenure and leasehold governance, tariff access, electricity/gas price ratios, and market pass-through allow the engineering change to reduce household exposure.

5.4 Figures and Tables

Expected tables will summarise baseline burden probabilities, modelled fuel-poverty diagnostics, price-shock burden uplift, Fuel Price-at-Risk, and intervention-specific de-risking dividends by income group, EPC rating, dwelling form, heating system, tenure, and LSOA/MSOA. The current high-risk explanation figures already promoted into the manuscript are Figures 1–6. Remaining expected figures will include: the conceptual distinction between fuel poverty, energy burden, and household fuel price risk; distributional plots of shock-induced burden uplift; intervention de-risking comparisons; and maps of de-risking gaps.

Before intervention results are reported, the spatial outputs should establish whether different mechanisms concentrate in different parts of Westminster. Priority maps include LSOA HR10 share, LSOA HR20 share, dominant HR10 pathway, social rented estate/block concentration, electric-heated inefficient flat concentration, and residual high-risk concentration. These maps would support the paper’s claim that fuel price risk is produced by locally specific combinations of housing form, income sensitivity, and governance constraints rather than by a uniform borough-wide condition.

The synched LSOA income PDF panels from the latest run are useful writing and diagnostic references. They show the estimated lognormal income distributions under selected λ values, with poverty-line and MSOA-mean markers. If promoted into the manuscript, they should be captioned as synthetic constrained income distributions for burden modelling, not as observed household income distributions and not as direct household fuel price risk outputs.

The paper will ultimately identify de-risking gaps: places and household-dwelling types where fuel price protection is most needed but low-carbon intervention is least accessible. These gaps are expected to be especially important where Westminster’s high housing costs, private renting, leasehold flats, heritage constraints, district-heating governance, and uneven tariff or asset access interrupt the translation from domestic net zero investment into household price-risk protection.

6 Discussion

The expected policy implication is that domestic net zero should be treated as a distributional household risk-protection agenda. Protection depends on the alignment of fabric retrofit, heating technology, distributed generation, flexibility, tariff design, electricity market reform, capital access, and tenure-sensitive governance.

In Westminster, this means paying particular attention to private renters, leasehold flats, HMOs, heritage-constrained dwellings, district-heating governance, social housing delivery pathways, and households whose vulnerability is masked by high neighbourhood average wealth.

The latest high-risk explanatory results sharpen this argument. Westminster’s current modelled high-burden tail is not produced by a single deficit such as low EPC performance alone. It emerges through multiple routes: high required heat demand, weak fabric, income sensitivity, tenure governance, electricity price exposure, and limited household control over building-level interventions. This matters because the policy question is not simply where average bills are high, but where future price shocks are likely to meet constrained adaptive capacity.

The interaction between dwelling size and EPC performance is especially important (Figure 2). Inefficient large dwellings have the highest within-type modelled burden probability, showing that poor fabric becomes most consequential where required energy demand is high. Yet the pathway typology also shows that intensity and scale are different policy problems (Figure 3). Inefficient large owner-occupied gas houses are a small but high-intensity group, while private rented weak-fabric gas flats and electric-heated inefficient flats or maisonettes contribute materially to the high-risk tail because they are common in the stock. A targeting strategy based only on the highest within-group risk would miss large numbers of exposed households; a strategy based only on stock counts would miss the households facing the steepest required-burden pressure.

Tenure changes the route through which de-risking can occur (Figure 4). Owner occupation does not automatically imply adaptive capacity where heritage rules, planning constraints, capital cost, and disruption limit retrofit. Social rented estate and block risk is primarily a delivery and governance problem: tenants cannot individually decide the timing, specification, or financing of estate-level fabric works, heating-system changes, or metering reform. Private rented weak-fabric flats raise a different problem of split incentives, enforcement, and minimum-standard compliance, especially where moderate aggregate EPC ratings obscure component-level fabric weakness.

The electric-heating pathway reinforces the paper’s caution about electrification (Figure 5). Electric-heated inefficient flats and maisonettes may already be exposed to high electricity unit costs. Unless fabric improvement, tariff reform, flexibility access, and electricity-market design reduce the household-facing price exposure, electrification can reduce emissions before it reduces household fuel price risk. The empirical strategy should therefore test clean heat, PV, batteries, smart tariffs, and integrated packages under explicit gas-to-electricity price-ratio assumptions rather than treating electrification as automatic protection.

The residual pathway is also substantively important. Nearly half of HR10 properties sit in the residual high-risk group, indicating that income sensitivity, local deprivation, and mixed mechanisms remain central to the current burden-risk distribution. The next modelling stage should therefore separate housing-driven, income-driven, and mixed-mechanism risk before estimating intervention de-risking dividends. This is not a weakness of the typology; it is evidence that the paper should avoid reducing household fuel price risk to a single housing-technology variable.

The broader implication is that domestic net zero policy must be evaluated against pathway-specific routes to protection. Fabric-first retrofit is most directly relevant where high required heat demand and poor energy performance dominate. Social housing pathways require block-level delivery, tenant protection, and post-intervention bill assessment. Private rented pathways require enforcement, landlord incentives, and tenant safeguards. Electric-heating pathways require tariff and flexibility reform as much as technology substitution. Residual income-driven pathways may need social tariffs, standing-charge reform, or income support before household risk can be materially reduced. The empirical test of domestic net zero is therefore not whether interventions look effective on average, but whether they reach the mechanisms through which exposure is produced.

7 Conclusion

Fuel poverty identifies current hardship; household fuel price risk identifies future exposure to price shocks. Net zero determines who receives protection only when domestic interventions are technically effective, financially accessible, and institutionally deliverable for the households most exposed.

The current Westminster evidence strengthens this framing. Modelled high-burden exposure is heterogeneous: large inefficient gas houses, social rented blocks, private rented weak-fabric flats, electric-heated inefficient flats and maisonettes, high-demand maisonettes, and residual income-sensitive cases represent different routes into risk and different routes out of it. A single technology narrative cannot explain this distribution.

The paper therefore asks not whether net zero reduces emissions, but whether, how, and for whom it credibly reduces household exposure to fossil-fuel price volatility. Its policy implication is correspondingly conditional: domestic net zero becomes household protection only where fabric, heating systems, tariffs, market design, finance, tenure, and local governance convert technical change into reduced upper-tail burden risk.